

## PART - B

### JAVA LAB PROGRAMS

1. A Cloth showroom has announced the following seasonal discount on purchase of items.

| Purchase Amount | Discount   |                |
|-----------------|------------|----------------|
|                 | Mill Cloth | Handloom items |
| 0 - 100         | -          | 5.0%           |
| 101 - 200       | 5.0%       | 7.5%           |
| 201 - 300       | 7.5%       | 10.0%          |
| Above 300       | 10.0%      | 15.0%          |

Write a program using switch and if statements to compute the net amount to be paid by a customer.

```
import java.io.*;
class Showroom
{
    public static void main(String args[]) throws IOException
    {
        InputStreamReader isr = new InputStreamReader(System.in);
        BufferedReader br = new BufferedReader(isr);

        System.out.println("1. For Mill Cloth");
        System.out.println("2. For Handloom items");

        System.out.println("Enter your choice...");
        int ch = Integer.parseInt(br.readLine());

        System.out.println("Enter the purchase amount...");
        int amount = Integer.parseInt(br.readLine());

        double discount = 0, netAmount = 0;

        switch(ch)
        {
```

```

case 1:
    if (amount <= 100)
        discount = 0;
    if (amount >=101 && amount <=200)
        discount = amount * 5.0 / 100;
    if (amount >=201 && amount <=300)
        discount = amount * 7.5 / 100;
    if (amount > 300)
        discount = amount * 10.0 / 100;

    netAmount = amount - discount;
    System.out.println("Purchase Amount: " + amount);
    System.out.println("Discount: " + discount);
    System.out.println("Amount paid by the customer: " +
netAmount);

    break;

case 2:
    if (amount <= 100)
        discount = amount * 5.0 / 100;
    if (amount >=101 && amount <=200)
        discount = amount * 7.5 / 100;
    if (amount >=201 && amount <=300)
        discount = amount * 10.0 / 100;
    if (amount > 300)
        discount = amount * 15.0 / 100;

    netAmount = amount - discount;
    System.out.println("Purchase Amount: " + amount);
    System.out.println("Discount: " + discount);
    System.out.println("Amount paid by the customer: " +
netAmount);

    break;

default:
    System.out.println("Invalid choice");
}
}
}

```

### Output:

```

javac Showroom.java
java Showroom

```

```
1. For Mill Cloth
2. For Handloom items
Enter your choice...
2
Enter the purchase amount...
25000
Purchase Amount: 25000
Discount: 3750.0
Amount paid by the customer: 21250.0
```

```
java Showroom
1. For Mill Cloth
2. For Handloom items
Enter your choice...
1
Enter the purchase amount...
15000
Purchase Amount: 15000
Discount: 1500.0
Amount paid by the customer: 13500.0
```

## 2. Write a program to find first N values of Fibonacci Series.

```
import java.io.*;
class Fibonacci
{
    public static void main(String args[]) throws IOException
    {
        InputStreamReader isr = new InputStreamReader(System.in);
        BufferedReader br = new BufferedReader(isr);

        System.out.println("Enter the N value for Fibonacci series...");
        int n = Integer.parseInt(br.readLine());

        int a = 0, b = 1, c;
        System.out.println("Fibonacci series are:");
        System.out.print(a + " " + b + " ");

        int i = 3;
        do
        {
```

```

        c = a + b;
        System.out.print(c + " ");
        a = b;
        b = c;
        i++;
    }
    while(i <= n);
}
}

```

### Output:

```

javac Fibonacci.java
java Fibonacci
Enter the N value for Fibonacci series...
10
Fibonacci series are:
0 1 1 2 3 5 8 13 21 34

```

**3. Write a program to accept A and B matrices and produce the third matrix  $C = A \times B$ . (Multiplication or Product of Two Matrix)**

```

import java.io.*;
class Product
{
    public static void main(String args[]) throws IOException
    {
        InputStreamReader isr = new InputStreamReader(System.in);
        BufferedReader br = new BufferedReader(isr);

        int a[][] = new int[10][10];
        int b[][] = new int[10][10];
        int c[][] = new int[10][10];

        System.out.println("Enter the number of rows and columns for first
matrix:");
        int r1 = Integer.parseInt(br.readLine());
        int c1 = Integer.parseInt(br.readLine());
    }
}

```

```

        System.out.println("Enter the number of rows and columns for second
matrix:");
        int r2 = Integer.parseInt(br.readLine());
        int c2 = Integer.parseInt(br.readLine());

        if(c1 != r2)
        {
            System.out.println("Matrix multiplication is not possible.");
            System.exit(0);
        }

        System.out.println("Enter the elements of first matrix:");
        for(int i = 0; i < r1; i++)
        {
            for(int j = 0; j < c1; j++)
            {
                a[i][j] = Integer.parseInt(br.readLine());
            }
        }

        System.out.println("Enter the elements of second matrix:");
        for(int i = 0; i < r2; i++)
        {
            for(int j = 0; j < c2; j++)
            {
                b[i][j] = Integer.parseInt(br.readLine());
            }
        }

        // Perform matrix multiplication
        for(int i = 0; i < r1; i++)
        {
            for(int j = 0; j < c2; j++)
            {
                c[i][j] = 0;
                for(int k = 0; k < c1; k++)
                {
                    c[i][j] += a[i][k] * b[k][j];
                }
            }
        }

        // Display the result
        System.out.println("Resultant matrix:");
        for(int i = 0; i < r1; i++)

```

```

    {
        for(int j = 0; j < c2; j++)
        {
            System.out.print(c[i][j] + " ");
        }
        System.out.println();
    }
}

```

### Output:

```

javac Product.java
java Product
Enter the number of rows and columns for first matrix:
2
3
Enter the number of rows and columns for second matrix:
3
2
Enter the elements of first matrix:
3
3
3
3
3
3
3
Enter the elements of second matrix:
4
4
4
4
4
Resultant matrix:
36 36
36 36

```

```

java Product
Enter the number of rows and columns for first matrix:
3
2
Enter the number of rows and columns for second matrix:
4
2

```

Matrix multiplication is not possible.

#### 4. Write a program to accept N String. Sort them in Ascending Order. (Alphabetical Order)

```
import java.io.*;
class Alphabetical
{
    public static void main(String args[]) throws IOException
    {
        InputStreamReader isr = new InputStreamReader(System.in);
        BufferedReader br = new BufferedReader(isr);

        String str[] = new String[10];
        System.out.println("Enter total number of strings:");
        int n = Integer.parseInt(br.readLine());

        System.out.println("Enter the strings:");
        for(int i = 0; i < n; i++)
        {
            str[i] = br.readLine();
        }

        // Sort the strings in alphabetical order
        for(int i = 0; i < n - 1; i++)
        {
            for(int j = i + 1; j < n; j++)
            {
                if(str[i].compareTo(str[j]) > 0)
                {
                    String temp = str[i];
                    str[i] = str[j];
                    str[j] = temp;
                }
            }
        }

        System.out.println("Strings in alphabetical order:");
        for(int i = 0; i < n; i++)
        {
            System.out.println(str[i]);
        }
    }
}
```

```
    }  
  }  
}
```

### Output:

```
javac Alphabetical.java  
java Alphabetical  
Enter total number of strings:  
5  
Enter the strings:  
hp  
sony  
lenovo  
acer  
dell  
Strings in alphabetical order:  
acer  
dell  
hp  
lenovo  
sony
```

5. Write a program to demonstrate usage of the classes and objects.

```
class Calculate  
{  
    int length, breadth;  
    void getData(int x, int y)  
    {  
        length = x;  
        breadth = y;  
    }  
    int calcArea()  
    {  
        int area = length * breadth;  
        return area;  
    }  
}
```

```

    }
}

class Rectangle
{
    public static void main(String args[])
    {
        Calculate c1 = new Calculate();
        c1.getData(10, 20);
        int area = c1.calcArea();
        System.out.println("Area of rectangle: " + area);

        Calculate c2 = new Calculate();
        c2.length = 5;
        c2.breadth = 10;
        int area2 = c2.length * c2.breadth;
        System.out.println("Area of rectangle: " + area2);
    }
}

```

#### Output:

```

javac Rectangle.java
java Rectangle
Area of rectangle: 200
Area of rectangle: 50

```

**6. Write a program to implement the concepts of multiple inheritance.**

```

class Student
{
    int rollno;
    void getNumber(int n)
    {
        rollno = n;
    }
    void displayNumber()
    {
        System.out.println("Roll Number = " + rollno);
    }
}

```

```

    }
}

class Test extends Student
{
    float part1, part2;
    void getMarks(float m1, float m2)
    {
        part1 = m1;
        part2 = m2;
    }
    void displayMarks()
    {
        System.out.println("Marks in Part 1 = " + part1);
        System.out.println("Marks in Part 2 = " + part2);
    }
}

interface Sports
{
    float sportwt = 6.0f;
    void displaySportswt();
}

class Result extends Test implements Sports
{
    float total;
    public void displaySportswt()
    {
        System.out.println("Sports Weightage = " + sportwt);
    }
    void displayTotal()
    {
        total = part1 + part2 + sportwt;
        displayNumber();
        displayMarks();
        displaySportswt();
        System.out.println("Total Score = " + total);
    }
}

class Multiple
{
    public static void main(String args[])
    {

```

```

        Result student1 = new Result();
        student1.getNumber(101);
        student1.getMarks(85.5f, 90.0f);
        student1.displayTotal();
    }
}

```

### Output:

```

javac Multiple.java
java Multiple
Roll Number = 101
Marks in Part 1 = 85.5
Marks in Part 2 = 90.0
Sports Weightage = 6.0
Total Score = 181.5

```

**7. Write a program to demonstrate the concepts of multithreaded programming.**

```

class A extends Thread
{
    public void run()
    {
        for (int i = 1; i <=5; i++)
        {
            if (i == 1)
                Thread.yield();
            System.out.println("\tFrom Thread A : i = " + i);
        }
        System.out.println("Exit from Thread A");
    }
}

class B extends Thread
{
    public void run()
    {
        for (int j = 1; j <=5; j++)
        {

```

```

        System.out.println("\tFrom Thread B : j = " + j);
        if (j == 3)
            break;
    }
    System.out.println("Exit from Thread B");
}
}

class C extends Thread
{
    public void run()
    {
        for (int k = 1; k <=5; k++)
        {
            System.out.println("\tFrom Thread C : k = " + k);
            if (k == 1)
                try
                {
                    Thread.sleep(1000);
                }
                catch (Exception e)
                {
                    System.out.println("Thread C interrupted");
                }
        }
        System.out.println("Exit from Thread C");
    }
}

class Multithread
{
    public static void main(String args[])
    {
        A threadA = new A();
        B threadB = new B();
        C threadC = new C();

        System.out.println("Starting Thread A");
        threadA.start();
        System.out.println("Starting Thread B");
        threadB.start();
        System.out.println("Starting Thread C");
        threadC.start();
        System.out.println("End of main thread");
    }
}

```

```
}
```

### Output:

```
javac Multithread.java
PS L:\JavaLabPrograms\Part-B> java Multithread
Starting Thread A
Starting Thread B
Starting Thread C
End of main thread
    From Thread A : i = 1
    From Thread C : k = 1
    From Thread B : j = 1
    From Thread A : i = 2
    From Thread A : i = 3
    From Thread A : i = 4
    From Thread B : j = 2
    From Thread B : j = 3
    From Thread A : i = 5
Exit from Thread A
Exit from Thread B
    From Thread C : k = 2
    From Thread C : k = 3
    From Thread C : k = 4
    From Thread C : k = 5
Exit from Thread C
```